Comparing Maintenance Policies for Wind Turbine Gearboxes Under Stochastic Degradation

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Abstract

Wind turbine gearboxes degrade under uncertain operating conditions, such as turbulence, storms, and rapidly changing loads, making maintenance decisions sensitive to randomness. While keeping turbines operating is crucial for energy production, poorly chosen maintenance strategies can lead to inadequate maintenance (higher failure risk and reduced output) or excessive maintenance (unnecessary cost and downtime). Inspired by previous work on this topic by Su et al., this paper uses a stochastic degradation model for the wind turbine gearbox's health to compare the efficacy of three maintenance approaches: condition-based maintenance, time-based incomplete maintenance, and a joint policy that combines the two. Deterministic analysis clarifies how each policy shapes the health trajectory, and stochastic simulations (Euler-Maruyama) show how noise changes threshold crossings and maintenance timing across parameter choices.

Keywords: Wind Turbines, Maintenance, Stochastic Differential Equations

1. Introduction

Wind energy capacity is rapidly increasing worldwide. Global Wind Energy Council (GWEC) reports that 2025 was a record year for new wind capacity and projects that the total global wind capacity will exceed 2 terawatts by 2030, roughly 1.5 times the total wind capacity at the end of 2024 [1, 2]. As wind deployment expands, the reliability of existing turbines becomes increasingly important. Maintaining high availability depends on preventing costly unplanned outages, especially for components with significant repair times.

A central challenge is the wind turbine gearbox, which must withstand humidity, temperature swings, and rapidly changing loads. Turbulence and fluctuating wind place significant stress on gears and bearings, which leads to damage and emergency shutdown [3]. Gearbox repair and reconditioning costs for a single turbine can be greater than $300,000, even after only a few years of operation [3]. This motivates policies that balance intervention cost against failure risk.

This paper looks at maintenance decisions through a stochastic degradation model for gearbox health. A purely time-based maintenance policy (TBM) is simple to implement but can over-maintain healthy gearboxes or under-maintain deteriorating ones. Condition-based maintenance (CBM) instead triggers action when health crosses a threshold, but in the field, the policy design is more nuanced than simply performing both types of maintenance. Threshold choices matter, and full replacement can be costly with large downtime, rendering it natural to include incomplete maintenance actions that improve the gearbox condition without restoring it to new. Su et al. [4] model this joint policy setting by incorporating incomplete maintenance alongside major maintenance and replacement, combining scheduled interventions with a condition-based trigger. Their work motivates viewing gearbox maintenance as a hybrid process, where the gearbox degrades continuously but is periodically reset or improved by discrete maintenance actions, and it shows that policy design can strongly affect long-run performance.

Inspired by [4], this paper models the gearbox condition as a degradation process with two maintenance actions: scheduled incomplete maintenance (TBIM) and threshold-triggered major maintenance (CBM). The guiding questions of this paper become: What maintenance settings keep gearbox health high while limiting how often we perform major maintenance? What are the trade-offs of each maintenance policy and how might that change in the random vs. deterministic model? We begin with the deterministic base case, then extend to qualitative stochastic analysis and simulation-based comparisons across policy parameters to address the question.

2. Model Overview

Let $x(t) \in [0, 1]$ denote gearbox health at time t, where $x(t) \approx 1$ indicates a healthy gearbox and values close to 0 indicate severe deterioration. Following the modeling approach in [4], we model gearbox health as a function

$x(t)$ that evolves over time, with randomness included to capture the unpredictable operating conditions (e.g. changing wind loads and storms).

Thus, the health of the gearbox is modeled by the stochastic differential equation:

$$dx(t) = -a\,x(t)\,dt + k\,x(t)\,dB(t). \tag{1}$$

The term $-a\,x(t)\,dt$ models the average deterioration of gearbox health over time, where $a > 0$ controls how quickly the decline occurs under typical conditions. The term $k\,x(t)\,dB(t)$ captures the unpredictable variation around this average trend caused by volatile operating conditions such as turbulence and shifting loads that are difficult to model deterministically. We represent the accumulation of many small disturbances with Brownian motion $B(t)$, which keeps the model tractable for analysis and simulation [5].

One limitation of this choice is that the noise term can occasionally push $x(t)$ upward over short intervals. We interpret these increases as fluctuations in the health index around the overall downward trend rather than physical self-repair. The model still trends downward on average, and increasing k makes the threshold-hitting time—and the resulting maintenance schedule—more variable.

The noise is assumed proportional to $x(t)$, consistent with the modeling approach in [4] and with the log-transform analysis used in Section 4.1. In our simulations, time t is measured in days with step size dt (we use $dt = 1$ day), and we convert time to years in plots by dividing by 365 for readability. Since $x(t)$ is dimensionless, a has units of $\frac{1}{\text{day}}$. The noise coefficient k has units of $\frac{1}{\sqrt{\text{day}}}$, reflecting that the variability coming from the Brownian motion term grows with the length of the time step.

2.1. Maintenance Policy

A key feature of the maintenance problem is that the gearbox health evolves continuously according to Eq. (1) between maintenance actions, but when maintenance occurs the state changes discretely (a jump in $x(t)$).

To define when maintenance occurs, we introduce the following thresholds and rules:

- **Condition-based maintenance (CBM):** We establish a threshold $X_c \in (0, 1)$ and define the major maintenance rule

  $$\text{if } \; x(t) \le X_c, \quad x \leftarrow 1.$$

  Thus, whenever the health reaches or falls below X_c, we immediately perform major maintenance and reset the gearbox health to 1.

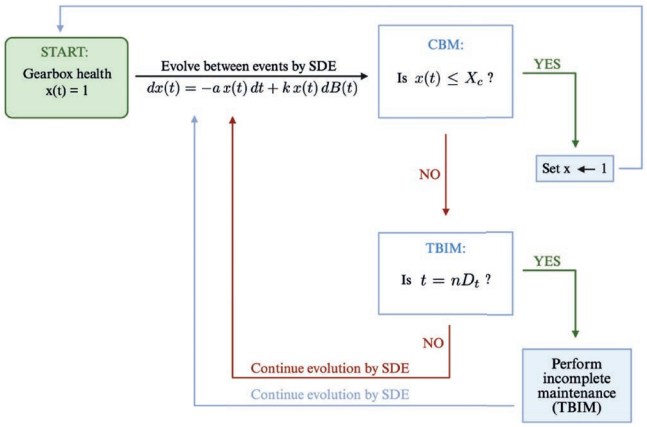


Figure 1: Model schematic of the joint TBIM + CBM maintenance logic. *Created with BioRender.com*

- **Time-based incomplete maintenance (TBIM):** We choose a time interval $D_t > 0$ such that maintenance occurs at times $t = nD_t$ for $n = 1, 2, 3, \ldots$. At each scheduled time, we apply the fractional improvement rule

  $$x \leftarrow x + \rho(1 - x), \qquad 0 < \rho < 1.$$

  This update increases the current health but generally does not return it to 1; the parameter ρ controls the strength of the maintenance action, with larger ρ meaning a larger fraction of the remaining gap $(1-x)$ to 1 is repaired.

Figure 1 summarizes the Eq. (1) policy logic; the gearbox health evolves by the SDE between events, with CBM resets at $x(t) \le X_c$ and scheduled TBIM updates at $t = nD_t$.

3. Deterministic Base Case

We first set $k = 0$ to understand how the maintenance rules behave when the gearbox health follows a single, predictable deterioration path. This baseline lets us compute key quantities directly and create reference plots that we will later compare to the stochastic model.

We begin with the stochastic degradation model

$$dx(t) = -a\,x(t)\,dt + k\,x(t)\,dB(t), \qquad a > 0,\ k \ge 0,$$

and consider the deterministic base case obtained by setting $k = 0$. This yields the linear ODE

$$\frac{dx}{dt} = -a\,x(t), \qquad a > 0, \tag{2}$$

which has the solution

$$x(t) = x_0 e^{-at}. \tag{3}$$

Parameter	Description	Values used
a	deterioration rate (per day)	0.001
x_0	initial gearbox health	1
X_c	CBM threshold	0.4 (baseline); 0.3, 0.4, 0.5
D_t	TBIM interval (days)	730 (baseline); 365, 730, 1095
ρ	TBIM strength	0.50
t_{end}	simulation duration (days)	10×365

Table 1: Deterministic model parameters used to generate the baseline policy plots.

If the gearbox is perfectly healthy at $t = 0$ such that $x_0 = 1$, then $x(t) = e^{-at}$. Eq. (3) shows that larger a corresponds to faster exponential deterioration.

Figures 2-4 that follow show the deterministic trajectories under each policy for a fixed parameter set in Table 1 chosen to make the maintenance events visible over a ten-year period.

3.1. Time-to-threshold for CBM

Recall $0 < X_c < 1$ denotes the CBM threshold. In the CBM-only setting, meaning no scheduled maintenance, CBM is triggered the first time the deterministic trajectory reaches X_c, so when

$$x(\tau_c) = X_c.$$

Thus,

$$\tau_c = \frac{1}{a} \ln\left(\frac{x_0}{X_c}\right), \qquad x_0 > X_c. \tag{4}$$

If $x_0 = 1$, then $\tau_c = \frac{1}{a} \ln\left(\frac{1}{X_c}\right)$. In the deterministic CBM-only case with resetting the gearbox health x to 1 at each trigger, τ_c represents the time between major maintenance events. Figure 2 illustrates the CBM-only trajectory for several X_c thresholds, showing how the threshold controls the reset timing.

As X_c increases, CBM events occur sooner and therefore more often over the same 10 year time horizon.

3.2. TBIM-only Health Dynamics

We next isolate the effect of scheduled incomplete maintenance by considering TBIM-only: maintenance performed at fixed time intervals with no CBM resets applied.

Let $D_t > 0$ denote the amount of time between scheduled maintenance, so scheduled times are

$$t = D_t,\ 2D_t,\ 3D_t,\ \ldots$$

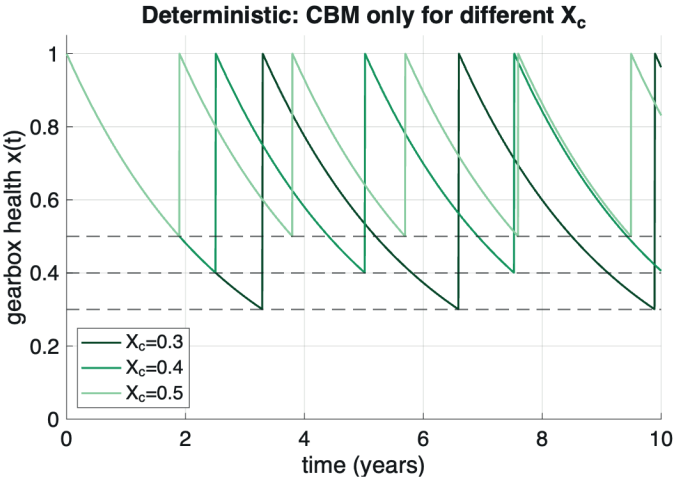


Figure 2: Gearbox health for different maintenance thresholds, X_c

To keep track of the gearbox health at those scheduled times, we define x_n to be the health at the n-th scheduled maintenance time:

$$x_n = x(nD_t), \qquad n = 0, 1, 2, \ldots,$$

where $x_0 = x(0) \in [0, 1]$ is the initial health.

Between scheduled times, the health follows the deterministic solution given by Eq. (3), so starting from x_n at time $t = nD_t$, the health at a later time $t \in [nD_t, (n+1)D_t]$ is

$$x(t) = x_n e^{-a(t-nD_t)},$$

and just before the next scheduled maintenance time $t = (n+1)D_t$, the health is

$$\begin{aligned} x\big((n+1)D_t\big) &= x_n e^{-a((n+1)D_t - nD_t)}, \\ &= x_n e^{-aD_t}. \end{aligned}$$

At time $(n+1)D_t$, we apply incomplete maintenance using the fractional improvement rule:

$$x \ \leftarrow \ x + \rho(1 - x), \qquad \rho \in (0, 1).$$

Applying this update to the pre-maintenance health $x_n e^{-aD_t}$ gives

$$x_{n+1} = x_n e^{-aD_t} + \rho\big(1 - x_n e^{-aD_t}\big) \tag{5}$$

$$= (1 - \rho)\, x_n e^{-aD_t} + \rho. \tag{6}$$

Under TBIM-only with the fractional repair rule, the health at scheduled maintenance times satisfies a simple step-by-step update. Each cycle consists of exponential decay over D_t, followed by a partial recovery that moves the health a fraction ρ of the way back toward 1. Because $0 < \rho < 1$ and $a > 0$, the decay factor over the interval

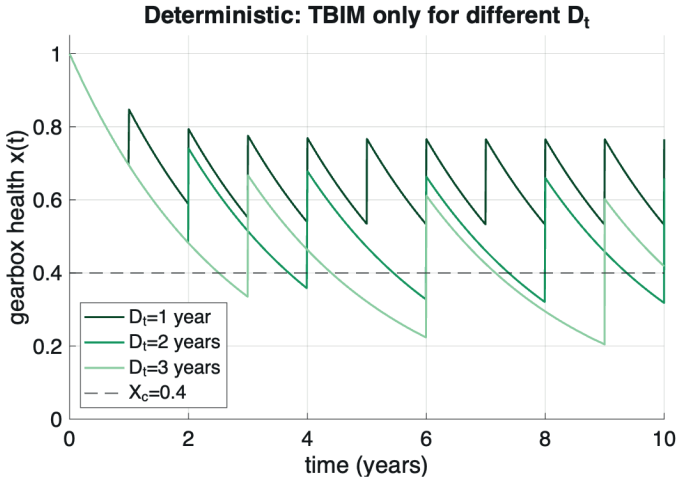


Figure 3: Incomplete maintenance every D_t with no CBM

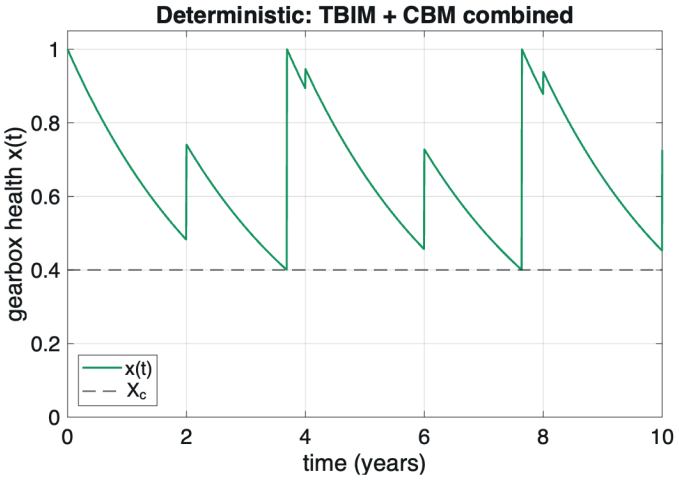


Figure 4: Combination of incomplete maintenance every D_t and CBM resets

is less than 1, so the effect of earlier values shrinks over time. Figure 3 shows TBIM-only trajectories for several scheduled maintenance interval lengths D_t, isolating how TBIM changes the decay pattern.

Larger D_t results in longer decay between repairs, so the minimum health reached before each TBIM event is lower.

3.3. Joint Policy: TBIM + CBM

We now combine the two maintenance rules: (i) scheduled incomplete maintenance performed every D_t units of time, and (ii) CBM triggered when $x(t) \leq X_c$. In the deterministic case, we have a piecewise trajectory with exponential decay between events and discrete jumps due to maintenance actions.

As in the TBIM-only case, we let

$$x_n = x(nD_t), \qquad n = 0, 1, 2, \ldots$$

denote the health at the scheduled maintenance times.

The CBM threshold X_c may be reached before the next scheduled maintenance. Since $x(t)$ is decreasing over $[nD_t, (n+1)D_t]$, CBM is triggered during this interval if and only if

$$x_n e^{-aD_t} \leq X_c. \tag{7}$$

If Eq. (7) holds, then CBM is triggered within the interval at the time t when $x(t) = X_c$. We obtain this t as follows:

$$\begin{aligned} x(t) = x_n e^{-a(t-nD_t)} &= X_c, \\ -a(t - nD_t) &= \ln\left(\frac{X_c}{x_n}\right), \\ t &= nD_t + \frac{1}{a}\ln\left(\frac{x_n}{X_c}\right). \end{aligned}$$

At this time t, major maintenance is performed and the gearbox health is reset to 1. This means that the deterministic trajectory gets cut off early by the CBM threshold before the next scheduled incomplete maintenance event can occur.

If instead $x_n e^{-aD_t} > X_c$, then CBM does not occur during $[nD_t, (n+1)D_t]$ and the trajectory reaches the next scheduled maintenance time. In that case, TBIM is applied at $t = (n+1)D_t$ using the fractional improvement rule,

$$x \;\leftarrow\; x + \rho(1 - x), \qquad \rho \in (0, 1),$$

which gives the same update as in the TBIM-only case:

$$x_{n+1} = (1 - \rho)x_n e^{-aD_t} + \rho.$$

In this deterministic combined-policy setting, the main consideration is whether the gearbox health drops far enough during one scheduled maintenance interval to reach the CBM threshold. Once we add randomness back into the model, some gearbox health trajectories will hit X_c earlier than predicted here, while others will hit later, changing both when major maintenance occurs and how often it happens over the long run. Figure 4 shows the combined policy, where scheduled TBIM events may delay, but not necessarily prevent, CBM events.

In this parameter setting, TBIM delays some threshold crossings, so fewer CBM events occur over the same time horizon compared to CBM-only at the same X_c.

3.4. Average Health Comparison: CBM vs. TBIM

To compare CBM and TBIM in the deterministic setting, we use the average gearbox health over time as a performance measure. This gives a single number that lets

us compare policies that act in different ways—CBM produces occasional full resets while TBIM produces smaller scheduled boosts—using one consistent way to measure how healthy the gearbox remains over the time horizon.

To find the average value of $x(t)$ over an interval $[t_0, t_1]$, we take the integral of x over $[t_0, t_1]$ and divide it by the interval's length as follows:

$$\overline{x} = \frac{1}{t_1 - t_0} \int_{t_0}^{t_1} x(t)\, dt.$$

3.4.1 CBM-only Average

In the CBM-only deterministic setting (with gearbox health resets to 1), one CBM interval starts at $x(0) = 1$ and ends when $x(\tau_c) = X_c$. We recall that over this interval,

$$x(t) = e^{-at}, \qquad \tau_c = \frac{1}{a} \ln\left(\frac{1}{X_c}\right).$$

Thus,

$$\overline{x}_{\text{CBM}} = \frac{1}{\tau_c} \int_0^{\tau_c} e^{-at}\, dt.$$

Evaluating the integral gives

$$\begin{aligned} \int_0^{\tau_c} e^{-at}\, dt &= \left[-\frac{1}{a} e^{-at}\right]_0^{\tau_c} \\ &= \frac{1}{a}\left(1 - e^{-a\tau_c}\right). \end{aligned}$$

Since $e^{-a\tau_c} = X_c$, this becomes

$$\int_0^{\tau_c} e^{-at}\, dt = \frac{1}{a}(1 - X_c).$$

Substituting back into our equation for $\overline{x}$ yields

$$\begin{aligned} \overline{x}_{\text{CBM}} &= \frac{\frac{1}{a}(1 - X_c)}{\frac{1}{a} \ln\left(\frac{1}{X_c}\right)} \\ &= \frac{1 - X_c}{\ln\left(\frac{1}{X_c}\right)}. \end{aligned}$$

Figure 5 summarizes the CBM-only tradeoff as the threshold X_c varies. Increasing X_c triggers major maintenance sooner, which raises the average health between CBM events (left axis). At the same time, earlier triggers shorten the interval between resets, so the expected number of CBM events over a fixed amount of time increases (right axis).

3.4.2 TBIM-only Average

To study TBIM-only in the deterministic setting, we ignore the CBM trigger and consider what happens when gearbox health decays deterministically between scheduled times and is partially repaired every D_t. Since

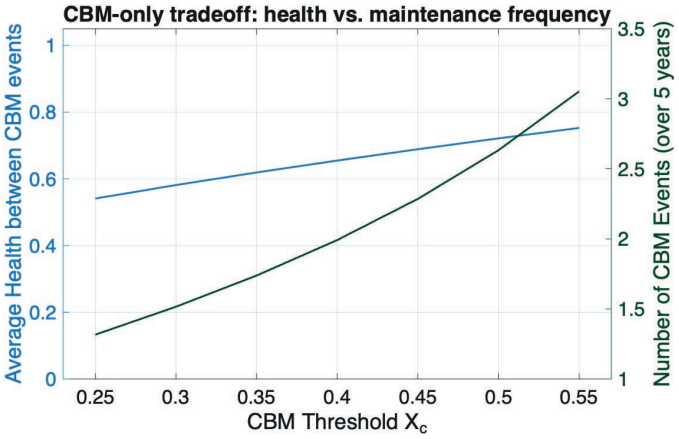


Figure 5: Average gearbox health and expected CBM resets over 5 years vs. threshold X_c

TBIM performance depends primarily on the maintenance interval D_t and the maintenance strength ρ, we summarize policy behavior across a grid of (D_t, ρ) values.

To compare TBIM-only behavior in a consistent way across different choices of D_t, we compute the average gearbox health over a fixed horizon $[0, T]$,

$$\bar{x}_{TBIM} = \frac{1}{T} \int_0^T x(t)\, dt.$$

For each (D_t, ρ) on the grid, we compute $\bar{x}_{TBIM}$ numerically by summing the area under the deterministic trajectory $x(t)$ on each interval between scheduled TBIM events. We choose T to be long enough that each parameter pair includes many scheduled TBIM events, with T taken as a fixed number of TBIM cycles. See Appendix for the calculation for a single interval and how we handle the final partial interval. Figure 6 shows the resulting heat map, where each pixel represents the computed value of $\bar{x}_{TBIM}$.

Figure 6 shows how average health under TBIM-only changes with maintenance frequency D_t and strength ρ. For a fixed maintenance interval D_t, increasing ρ increases $\bar{x}_{TBIM}$ because each maintenance action restores a larger fraction of the remaining damage. For a fixed maintenance strength ρ, increasing D_t decreases $\bar{x}_{TBIM}$ because the gearbox spends a longer period of time decaying before each incomplete maintenance event. The highest average health region occurs when maintenance is both frequent and effective, while long intervals combined with weak maintenance produce substantially lower average health over the same time horizon. In the joint-policy setting, parameter choices that already yield low $\bar{x}_{TBIM}$ under TBIM-only would require frequent CBM events once the threshold trigger is reintroduced, while parameter choices

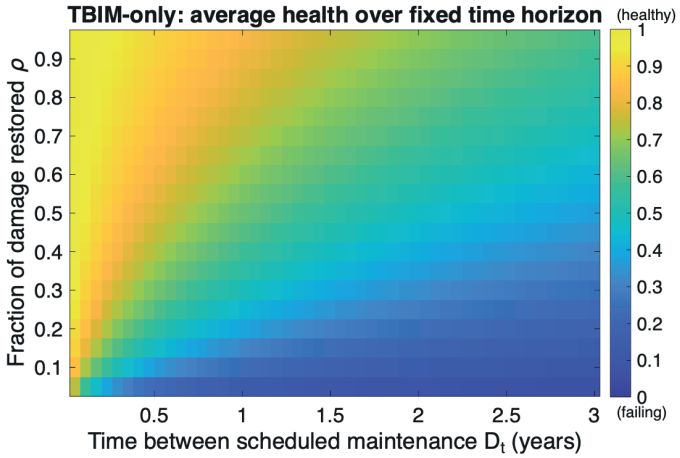


Figure 6: TBIM-only heat map of average health over $[0, T]$ across (D_t, ρ), with $T = 50 * \max(D_t)$

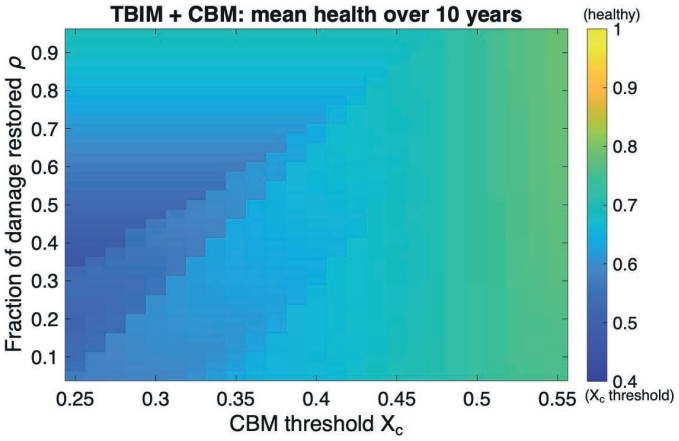


Figure 7: Mean gearbox health over 10 years under the deterministic joint TBIM + CBM policy, varying X_c and ρ with $D_t = 2$ years fixed

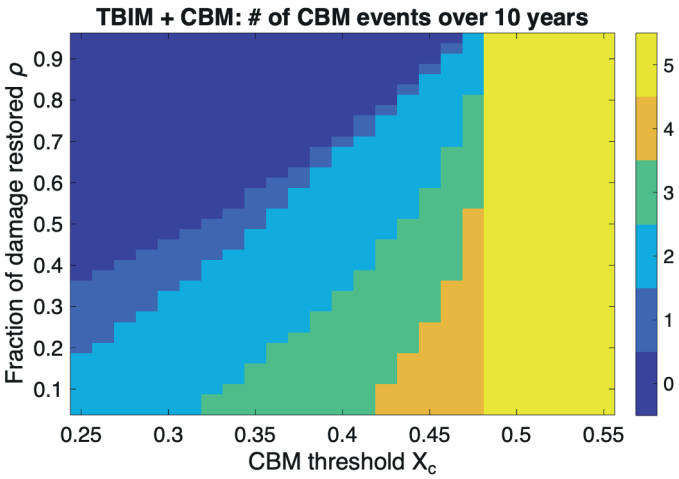


Figure 8: Number of CBM events over 10 years under the deterministic joint TBIM + CBM policy, varying X_c and ρ with $D_t = 2$ years fixed

that maintain high average health should correspond to few CBM events.

3.4.3 Policy Tradeoffs

These deterministic results clarify the main maintenance policy tradeoffs. In the CBM-only case, raising the threshold X_c increases the average health between resets, but also increases the number of major maintenance events, which is costly and causes downtime. In the TBIM-only case, average health over a fixed horizon depends on both the scheduled interval for incomplete maintenance D_t and the maintenance strength ρ. Together, these base case calculations indicate when scheduled incomplete maintenance is strong and frequent enough to keep the gearbox's health above the threshold X_c, and when it is not, so that CBM events would be required once the condition trigger is reintroduced. This motivates the combined policy as a balance between scheduled incomplete maintenance that reduces deterioration through regular partial repairs and CBM as a backstop that resets the gearbox when health falls too low.

We next visualize how the combined TBIM + CBM policy behaves as the CBM threshold X_c and incomplete-maintenance strength ρ vary, with the other parameters fixed as in Table 1. For each (X_c, ρ), we simulate the deterministic trajectory over ten years and record the mean gearbox health over the horizon and the number of CBM events performed. We use a ten-year horizon so that even at $D_t = 2$ years the model includes multiple scheduled TBIM actions, making differences across (X_c, ρ) easier to compare.

Figure 7 shows that increasing ρ raises the mean health over the ten-year period, while increasing X_c also tends to raise mean health because CBM is triggered earlier. Figure 8 shows the tradeoff behind that improvement; larger X_c increases the number of major maintenance events performed, while larger ρ reduces the need for CBM events by keeping the gearbox health from falling to the threshold between scheduled TBIM events. For higher CBM thresholds ($X_c \gtrsim 0.5$), CBM is triggered repeatedly over the ten-year horizon (five events), regardless of ρ, indicating that scheduled TBIM every $D_t = 2$ years cannot prevent full-reset maintenance events when the trigger level is set high. In contrast, for moderate and lower thresholds X_c, increasing ρ creates a large region where CBM never occurs, meaning TBIM alone is sufficient to keep the deterministic trajectory above X_c throughout the horizon.

These deterministic parameter maps motivate the stochastic analysis and simulations that follow. When noise is added, trajectories will cross the same thresholds earlier or later, but the same tradeoff between maintaining higher health and avoiding frequent major resets remains central.

4. Stochastic Model

In this section, we return to the full stochastic model Eq. (1) ($k > 0$), where gearbox health is no longer a single deterministic curve but a random process. We denote the health as $X(t)$ to emphasize that, with noise present, the health at a fixed time t can differ across realizations even when the parameters are the same.

Before moving to simulations, we derive a useful representation of $X(t)$ to clarify how the noise level k affects the dynamics. A key feature of our SDE is that the random term is proportional to the current health level so random fluctuations enter multiplicatively. A standard way to simplify multiplicative noise models is to apply a logarithmic change of variables. Thus, we let $Y(t) = \ln X(t)$. This converts multiplicative fluctuations in $X(t)$ into additive fluctuations in $Y(t)$, which makes it easier to see how k affects the trajectory.

To carry out this change of variables, we use Itô's formula [5] to compute the stochastic differential of $\ln X(t)$. Because $X(t)$ includes Brownian motion, this step replaces the usual chain rule and produces the correct differential for $\ln X(t)$.

4.1. Log Transform

Recall that $X(t)$ satisfies the stochastic degradation model

$$dX(t) = -a\,X(t)\,dt + k\,X(t)\,dB(t), \qquad a > 0,\ k \geq 0,$$

where $B(t)$ is standard Brownian motion. We apply Itô's formula as given in [5] to $f(x) = \ln x$, and define $Y(t) = \ln X(t)$. Carrying out the calculation (details in Appendix), we obtain an additive-noise SDE for $Y(t)$:

$$dY(t) = \left(-a - \frac{1}{2}k^2\right) dt + k\,dB(t). \tag{8}$$

In this form, the deterministic component $(-a-\frac{1}{2}k^2)dt$ describes the average downward trend in $\ln X(t)$, while the term $k\,dB(t)$ introduces random fluctuations whose size is controlled by k. This form is useful because it separates the model into a deterministic trend term and a noise term with constant diffusion coefficient k. Integrating Eq. 8 yields the explicit representation for $X(t)$:

$$X(t) = x_0\, e^{\left(-a-\frac{1}{2}k^2\right)t + kB(t)}. \tag{9}$$

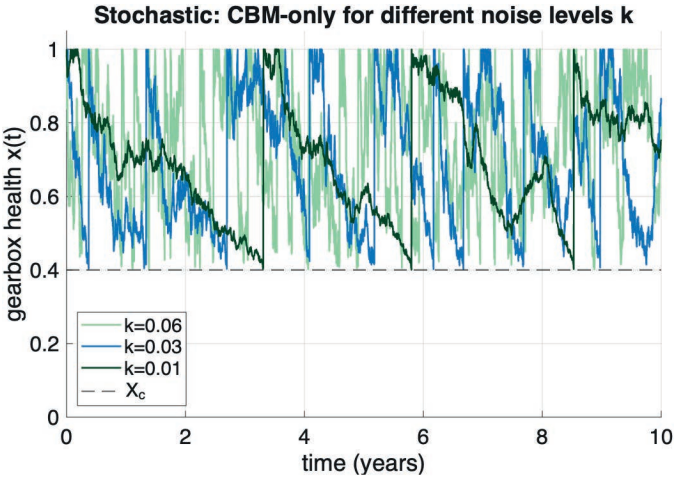


Figure 9: Sample trajectories of the stochastic CBM-only policy over ten years for three noise levels k

In the deterministic case where $k = 0$, the health trajectory is $x(t) = x_0 e^{-at}$, so threshold-hitting times and maintenance triggers are fully determined by the choice of parameters. In the stochastic case where $k > 0$, the solution Eq. (9) depends on Brownian motion and is thus random, so the time at which $X(t)$ first hits a threshold becomes a random variable. This difference is central for condition-based policies, as even with the same parameters, different realizations of the degradation process can trigger maintenance earlier or later. Increasing k increases the variability of potential paths, which widens the spread of threshold-hitting times and resulting maintenance schedules. These observations motivate the numerical simulations run in the next section, where we implement the maintenance rules and estimate long-run behavior across parameter choices.

4.2. Simulations

We simulate the SDE model Eq. (1) using the Euler-Maruyama scheme [6], and apply the CBM and TBIM maintenance rules whenever their conditions are met. To visualize how noise changes the degradation behavior, we first plot representative trajectories of the stochastic model under the two single-policy settings CBM-only and TBIM-only. All parameters are the same as in Table 1 and we introduce X_{low} to denote gearbox health being below an even more critical threshold of 0.2. We hold $(a, D_t, X_c, \rho, X_{\text{low}})$ fixed and vary only the noise level $k \in [0.01, 0.03, 0.06]$ to isolate the effect of randomness. After each Euler-Maruyama update and maintenance jump, we cap the health variable to remain in $[0, 1]$ by setting values above 1 to 1 and values below 0 to 0, consistent with our definition of the gearbox health index.

Because CBM events reset the gearbox to 1 whenever $X(t) \leq X_c$, the trajectories in Figure 9 remain bounded

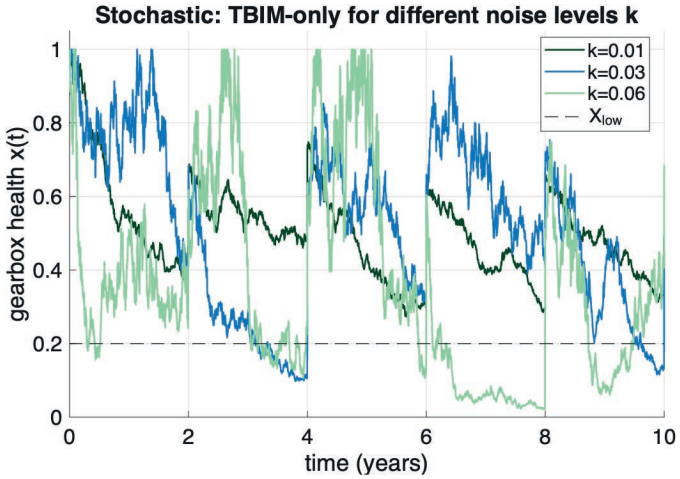


Figure 10: Sample trajectories of the stochastic TBIM-only policy over ten years for three noise levels k

away from low health. However, as k increases, trajectories become more irregular and hit the threshold X_c more unpredictably, leading to more frequent resets over the same time period, most prominently shown by the light green trajectory for $k = 0.06$. We recognize a key distinction from the deterministic case; under randomness, the CBM trigger time is not exclusively determined by the model's parameters, increasing k increases variability in maintenance timing.

Figure 10 shows sample paths under the TBIM-only policy for the same three values of k. As k increases, the trajectories in Figure 10 exhibit noticeably larger fluctuations around the baseline deterioration trend. In particular, we see that higher noise produces more frequent and longer excursions below X_{low}. This supports using time spent in low health as a performance metric in the stochastic setting as even with an identical TBIM policy (D_t, ρ), larger k increases the likelihood that the gearbox enters undesirable health states between TBIM events.

While Figures 9-10 help visualize individual realizations of the stochastic model, single trajectories are not representative of long-run performance. Therefore, we estimate performance metrics by averaging over many trajectories. For each policy and each k, we generate 50 independent sample paths over a ten year period and report the mean number of CBM events, mean number of TBIM events, the average health of the gearbox $\bar{x}$, and the percent of time the gearbox spends below the low-health threshold X_{low}. The MATLAB implementation was informed by the Euler-Maruyama template in [7].

Table 2 shows that increasing the noise level k substantially worsens the no-maintenance baseline as mean health

Table 2: Stochastic simulation results (means over 50 runs, ten-year horizon). Columns report the mean number of CBM events, mean number of TBIM events, mean gearbox health $\bar{x}$, and percent of time spent with the gearbox in low health (below X_{low}).

k	**Policy**	**CBM**	**TBIM**	$\bar{x}$	**% time** $< X_{\text{low}}$
0.01	No maint.	0.0	0.0	0.256	58.0
0.01	CBM only	4.1	0.0	0.675	0.0
0.01	TBIM only	0.0	5.0	0.522	0.6
0.01	CBM + TBIM	3.0	5.0	0.687	0.0
0.03	No maint.	0.0	0.0	0.185	70.0
0.03	CBM only	7.5	0.0	0.719	0.0
0.03	TBIM only	0.0	5.0	0.451	15.7
0.03	CBM + TBIM	7.5	5.0	0.725	0.0
0.06	No maint.	0.0	0.0	0.093	85.5
0.06	CBM only	21.9	0.0	0.737	0.0
0.06	TBIM only	0.0	5.0	0.310	45.4
0.06	CBM + TBIM	19.9	5.0	0.745	0.0

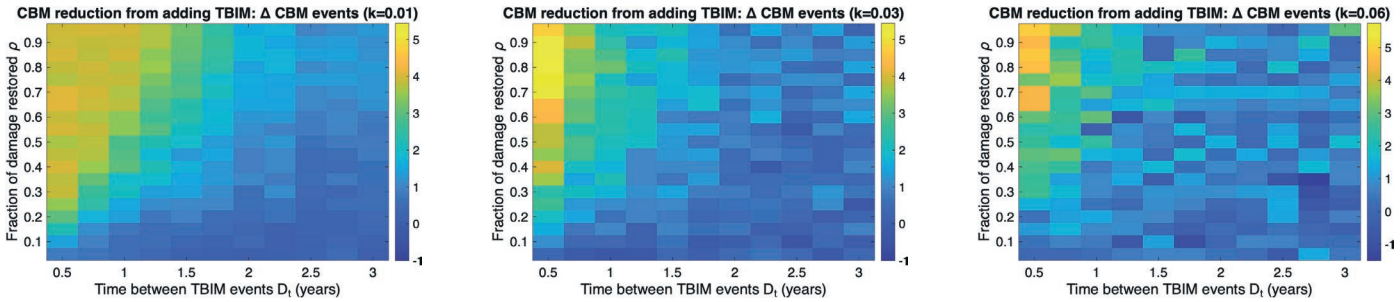
$\bar{x}$ decreases from 0.256 to 0.093 and the percent of time spent below X_{low} rises from 58.0% to 85.5% as k increases from 0.01 to 0.06. Under CBM-only, the system spends no time below X_{low} for all k given $X_c > X_{\text{low}}$, but the mean number of CBM events increases from 4.1 to 21.9, reflecting more frequent threshold crossings under higher variability. Under TBIM-only with the fixed (D_t, ρ) values in Table 1, performance degrades with k as TBIM events remain fixed by the schedule while time in low-health increases from 0.6% to 45.4%. The combined CBM + TBIM policy keeps health above X_{low} for all k given the CBM threshold, but like CBM-only requires many CBM events when k is large.

Table 2 compares policies at a single parameter set, which hides when TBIM meaningfully changes CBM behavior in the joint policy setting. Since the joint policy adds cost through scheduled TBIM on top of major CBM events, our key question is whether TBIM can reduce the frequency of CBM events, and for what parameter choices. To isolate this effect, we fix the CBM threshold X_c as in Table 1 and examine a grid of TBIM parameters (D_t, ρ). For each noise level $k \in [0.01, 0.03, 0.06]$, we estimate the mean number of CBM events under CBM-only and under the joint policy, and plot their difference.

Let $\bar{N}$ be the mean number of CBM events over n_s simulated paths. Let $\bar{N}_{\text{CBM-only}}$ denote this mean under the CBM-only policy, and let $\bar{N}_{\text{CBM+TBIM}}$ denote the corresponding mean under the joint policy with TBIM parameters (D_t, ρ). We define

$$\Delta_{\text{CBM}}(D_t, \rho) = \bar{N}_{\text{CBM-only}} - \bar{N}_{\text{CBM+TBIM}}(D_t, \rho). \quad (10)$$

Positive values of Δ_{CBM} indicate that adding TBIM re-



(a) k = 0.01 (low noise) (b) k = 0.03 (moderate noise) (c) k = 0.06 (high noise)

Figure 11: Heat maps of the change in CBM frequency from adding TBIM, given by Eq. (10), over a (D_t, ρ) grid with X_c fixed. Subfigures (a)-(c) show results for $k = 0.01, 0.03, 0.06$, averaged over 50 simulated paths over a ten-year horizon.

duces the average number of CBM events, while negative values indicate that it increases them.

Figures 11a-11c summarize when TBIM works as a substitute for major maintenance events. Across all three noise levels, the clearest reductions in CBM events (positive Δ) occur when TBIM is strong enough (larger ρ) and frequent enough (smaller D_t) to keep gearbox health in sample paths from drifting down to $X = X_c$ between scheduled maintenance events. As D_t increases, TBIM becomes too infrequent to catch downward movements caused by noise, and Δ shrinks toward zero, meaning CBM is doing essentially all of the work maintaining the gearbox's health.

The effect also weakens as k increases; at higher noise, large downward health fluctuations can reach X_c quickly even if TBIM is reasonably strong, so TBIM prevents fewer CBM triggers on average. The patchier appearance at large k in Figure 11c reflects the sampling variability, but the overall pattern remains the same, with TBIM being most helpful in the region where it is both frequent and effective.

5. Discussion

Although this paper does not optimize an explicit cost function, the motivation for TBIM is operational as major gearbox interventions are expensive and can require substantial downtime. This matters in practice because gearbox repair and reconditioning can exceed $\$300,000$, replacement can cost up to $\$600,000$, and unplanned outages reduce turbine availability [3, 8]. In our simulations, Table 2 shows that when noise is moderate to large, CBM dominates performance in the sense that the joint policy often still triggers CBM at nearly the CBM-only rate, so TBIM adds scheduled interventions without substantially reducing major resets at the fixed parameter set. However, the heat maps in Figure 11 show that TBIM can act as a substitute for some CBM events in the stochastic setting when it is both frequent enough and strong enough to prevent trajectories from drifting down to X_c between scheduled visits. As D_t increases, Δ_{CBM} shrinks toward zero, indicating that TBIM becomes too infrequent to meaningfully change CBM behavior, and this effect weakens further as k increases.

A central limitation is that Brownian noise can produce short-term upward fluctuations in the gearbox health index, which should be interpreted as variability around the overall degradation trend rather than the gearbox physically repairing itself. Additionally, the drift-diffusion form used here is intentionally simple, with a constant deterioration rate and proportional noise. This simplicity keeps the policy effects transparent, so differences in CBM frequency and threshold crossings can be attributed to the maintenance rules and noise level rather than to additional modeling choices.

Further work might connect the health index to turbine performance outcomes (e.g., power production or efficiency loss), so that time spent at low health could be interpreted in terms of expected energy loss rather than a health statistic. Another extension would be to incorporate explicit downtime and cost terms, following approaches such as [4] that model maintenance-time damages and distinguish replacement maintenance from incomplete maintenance, so that policy comparisons can be made in terms of expected total cost and availability rather than event counts alone.

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CAROLINE HOFF



I used to think math was mostly a race. As many young students do, in elementary school we did "minute math," and I remember the specific kind of stress it gave me: the feeling that my ability was being measured by speed. I could do the problems, I had been adding and multiplying for years, but not fast enough to feel safe. I had an urge to finish before I even understood what I was doing. Even today, a lot of my relationship with math has been about time: whether I have enough of it, whether I'm using it well, and whether taking longer means I don't belong or am not useful in the field.

By the time I got to high school, I was placed on the accelerated track. What was supposed to be an exciting opportunity began to feel a bit like a punishment as I watched girls disappear from my classes until I found myself in rooms that were almost entirely boys. This punishment also became a banishment from the front row and a habit of leaving my hand down when I had a question so I wouldn't slow down the fast-paced class. This experience shifted when I ended up in a course with Mrs. Herabat, a former engineer who commanded the room. No side chatter or performative confidence, just a class that actually did the work. She spoke about her experiences managing infrastructure in exclusively male spaces, and how that experience pushed her toward teaching, because she wanted math to feel more available to more people. As the only girl in her class, I felt a strange kind of pressure to not waste the opportunity she described.

She also taught in a way that exposed one of my weak spots. She expected memorization for exams; we'd never be given the unit circle on a test, we were expected to have it in our brains. At some point I admitted to her that brute memorization didn't work for me. Instead of sympathy or a shrug, she said, "Then you must not be approaching it the right way. There is always a way to do anything that will work for you." I think about that advice more than I think about any trigonometric identity. It gave me a direction to stop treating the standard way as the only way. That idea has followed me through the rest of my math education: I don't always get things right away, but I'll stay after and work on my own until it clicks. My path isn't a straight climb, it's switchbacks. I double back when I need to and try not to apologize for taking a longer route.

I was excited to find myself in a linear algebra classroom during my first semester of college, and even more excited when it made me confident enough to pursue a math major. It made math feel connected to the world: eigenvalues and eigenvectors showing up in ranking and search made math feel like useful ideas with real consequences. Two semesters later, however, multivariable calculus shook that confidence. I felt unmoored, mostly because everything lived in higher dimensions that I struggled to draw or even picture. The material felt untethered, like it was happening in space, not on any terrain I could walk on. At the same time, I was taking differential equations, and the contrast was immediate. Differential equations felt tied to real processes–cooling and heating, fluid flow, systems changing over time–so even when the math was difficult, I could picture what it was trying to describe. That connection pushed me toward the applied math track: math that stays anchored to something I can understand.

Still, I didn't want to treat multivariable calculus as a closed door. Two semesters later, I audited it again. The biggest difference wasn't that it became easy, it didn't, but that I wasn't time-pressed anymore. I had a whole semester to sit with the ideas and take the switchbacks without the pressure of constant evaluation. I stopped equating slow with incapable and proved to myself that I would come back. That has reshaped what "useful" means to me in math. I used to think usefulness was quick understanding, quick solutions, quick confidence. Now I think it can also mean being careful and thorough, not rushing past confusion, and being able to explain ideas clearly to anyone. A lot of my imposter syndrome has come from reading math writing that feels like it assumes you already know the trail, which steps are safe to skip, and how to overtake its obstacles. I don't want my senior project to feel like that. I want it to be readable without pretending it's simple.